MOHAN KUMAR M

Software Engineer | B.E. Electronics & Communication Engineering

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OBJECTIVE

Final-year ECE student skilled in C, Python and full-stack web development (MERN, Next.js), with hands-on experience in IoT and AI. Built and deployed full-stack applications, including a live IoT platform and a freelance web service for real clients. Seeking a Software Engineer role to build reliable, user-focused applications.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Java
- **Frontend:** HTML, CSS (basics), JavaScript, responsive and mobile-first design, SEO
- **Backend & APIs:** Node.js(basics), Express.js(basics), REST API design and integration
- **Databases:** MongoDB, MySQL
- **Tools & Deployment:** Git/GitHub, Verel, Render, Netlify, Cloudinary
- **Core Concepts:** Object-Oriented Programming, Data Structures & Problem Solving

PROJECTS

Smart Hygiene Management System — *React.js, Node.js, Express.js, MongoDB, REST API, ESP32/C++*

- **Situation & Task:** Public waste bins lacked remote monitoring and secure access control, so I built a full-stack IoT platform to monitor and control smart waste bins remotely with authenticated multi-mode access.
- **Action:** Developed a React.js admin dashboard for real-time monitoring and remote unlock control; built Node.js/Express REST APIs with API-key-authenticated telemetry ingestion; designed the MongoDB schema with polling and auto-reconnect logic for reliable device-to-cloud data sync.
- **Result:** Delivered a live full-stack IoT application with three authenticated access modes (touch, RFID, remote admin) and real-time cloud-based control, deployed on Vercel and Render.

Mohan's WebCraft — Freelance Web Development Platform — *Next.js, Three.js, Vercel*

- **Situation/Task:** Built and launched a live web-development service platform delivering fast, mobile-first single-page sites for small businesses.
- **Action:** Built the front end in Next.js with interactive Three.js visuals, structured pricing tiers and a WhatsApp-based client intake flow; shipped on Vercel with SEO-optimized metadata.
- **Result:** Delivered a live, production-deployed site and onboarded freelance clients, gaining real-world experience turning client requirements into polished web interfaces.

INTERNSHIP EXPERIENCE

AI & ML Intern | Code Bind Technologies, Chennai | *Apr 2026 – May 2026*

- Developed a Hand Gesture Detection System using Python, OpenCV, and MediaPipe to track hand landmark coordinates and detect finger count in real time, gaining hands-on exposure to machine learning, computer vision, and AI application development.

EDUCATION

- **B.E., Electronics & Communication Engineering —** Government College of Engineering, Salem | 2023–2027 | CGPA: 7.54(till Sem 6)
- **HSC —**Equitas Gurukul Matriculation Higher Secondary School | 2022–2023 | 92.3%
- **SSLC —**Equitas Gurukul Matriculation Higher Secondary School | 2020–2021 | 100%

CERTIFICATIONS & ACTIVITIES

- **Smart India Hackathon (2025):** Contributed to problem-statement analysis and solution prototyping in a national-level innovation challenge.
- **Embedded Systems & IoT Workshop (2025):** Hands-on training in embedded architecture and IoT communication protocols.
- **Leadership & Interests:** Class Representative (1st year); Table Tennis player, Chess player, Sports Captain(During Schooling).